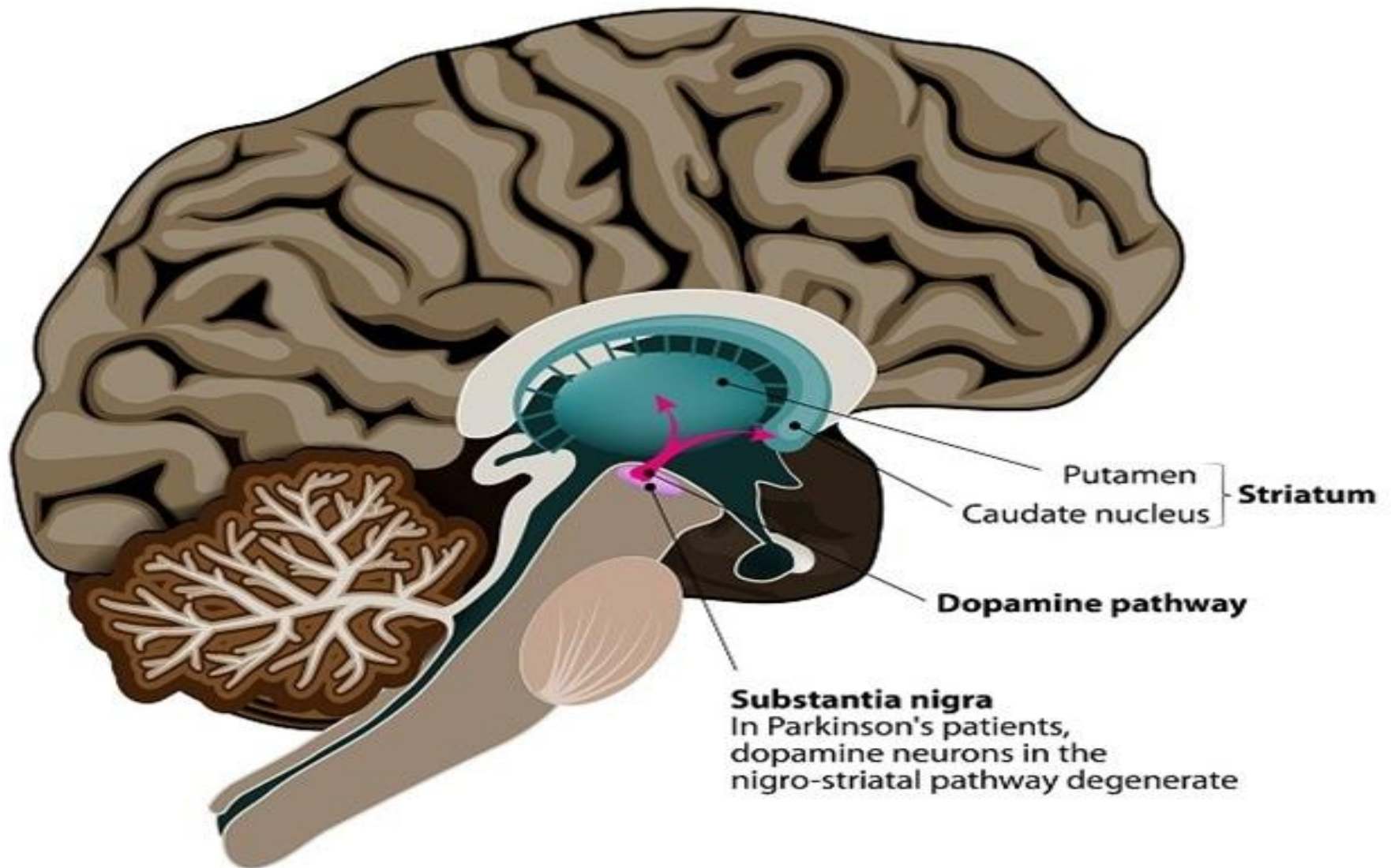


Drugs Used In Parkinsonism Disease

PARKINSON'S DISEASE



Parkinson's Disease



⇒ Signs

- ⇒ Akinesia (RT, initiation)
- ⇒ Bradykinesia (MT, slow)
- ⇒ Rigidity
- ⇒ Tremor
- ⇒ Postural instability, difficulty multi-tasking, sleep difficulties, cognitive deficits.

⇒ Cause

- ⇒ Denervation of DAergic neurons in SNpc
- ⇒ Loss of DA in striatum



Cut section
of the midbrain
where a portion
of the substantia
nigra is visible



Substantia nigra



Diminished substantia
nigra as seen in
Parkinson's disease



ADAM

○ **Parkinsonism = ↓ Dopamine in the nigrostriatal pathway**

Main clinical features:

- **Tremor**
- **Rigidity**
- **Bradykinesia**
- Postural/gait abnormalities
- Treatment basic aim:
- **Dopamine effect ↑ OR cholinergic effect ↓**



Parkinson's Disease



Rigidity



Tremor

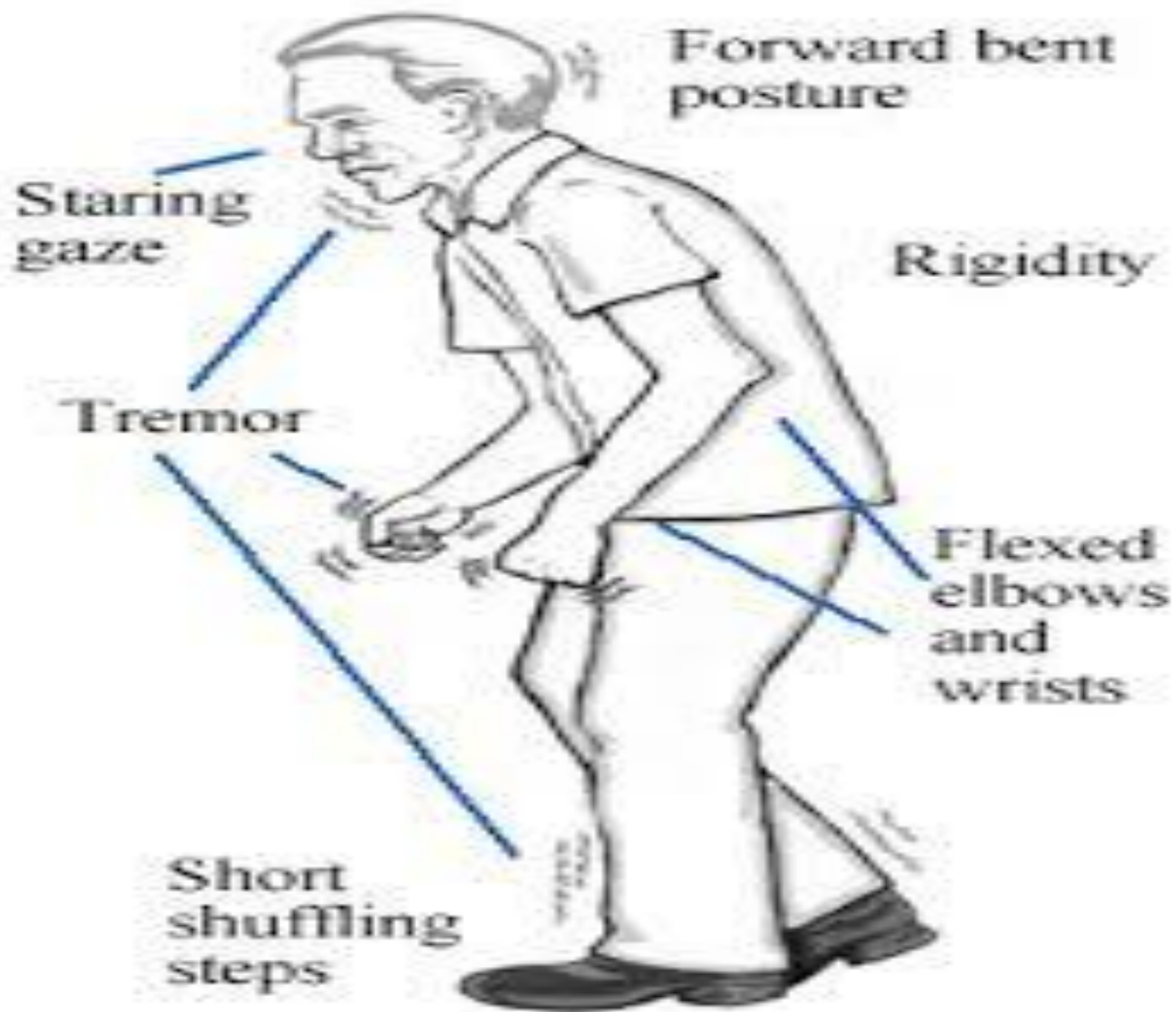


Postural instability

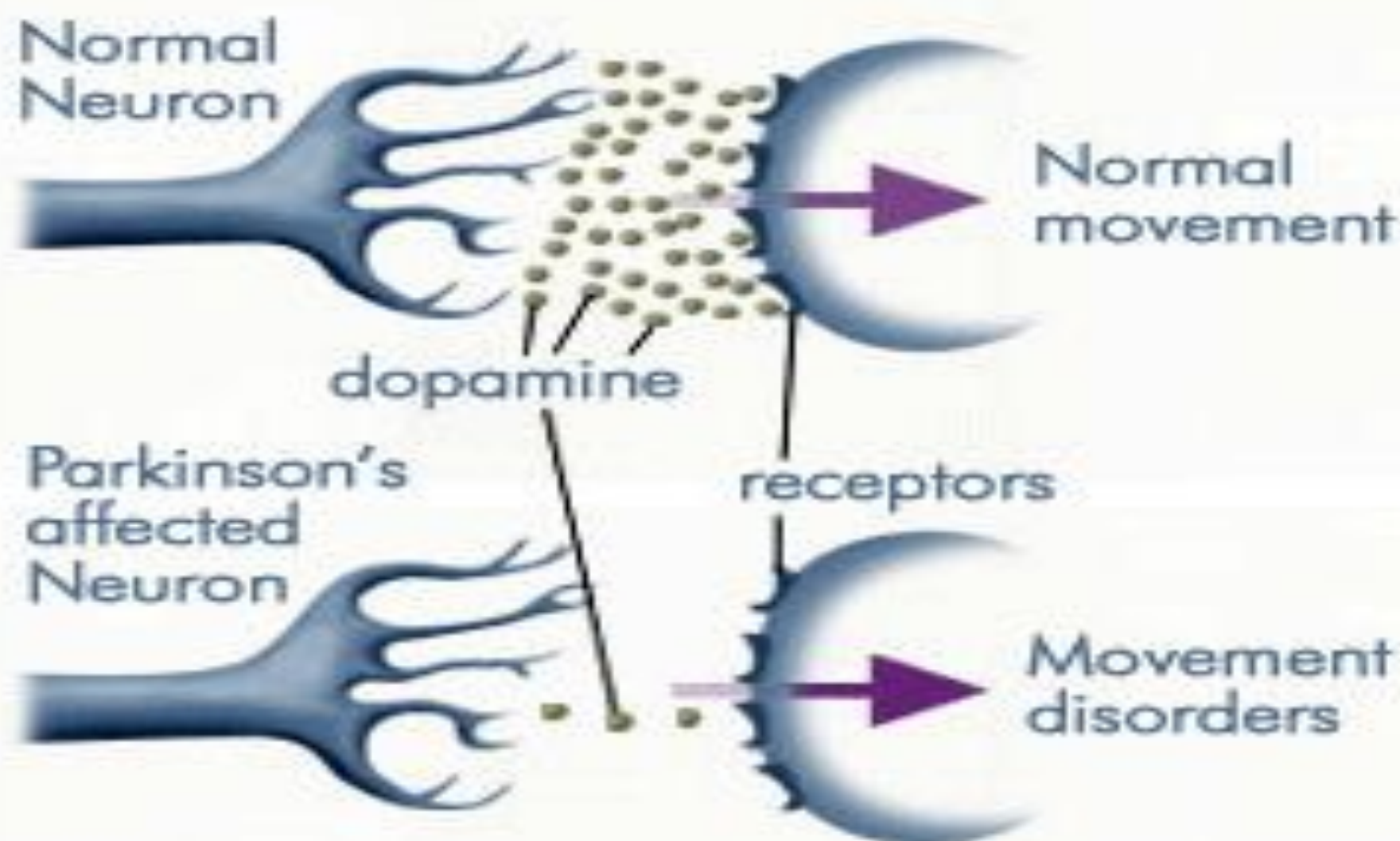


Bradykinesia





Dopamine levels in a normal and a Parkinson's affected neuron.



Dopamine ↓ → movement ↓

A. Dopamine ↑

Levodopa

Dopamine agonists

MAO-B inhibitors

COMT inhibitors

Amantadine

B. Acetylcholine effect ↓

Antimuscarinics

CLASSIFICATION OF ANTIPARKINSONISM DRUGS

1) Levodopa

2) Carbidopa

3) Dopamine Receptor Agonist

- Bromocriptine

- Pergolide

- Pramipexole

- Rotigotine

- Ropinirole

4) Apomorphine

5) Amantadine (Parenterally)

6) Monoamine Oxidase Inhibitors

- Selegiline - Rasagiline

**7) Catechole -O- Methyltransferase
(COMT) Inhibitors**

- Tolcapone - Tecapone

8) Antimuscarinic Agents

- Benztropine - Biperiden

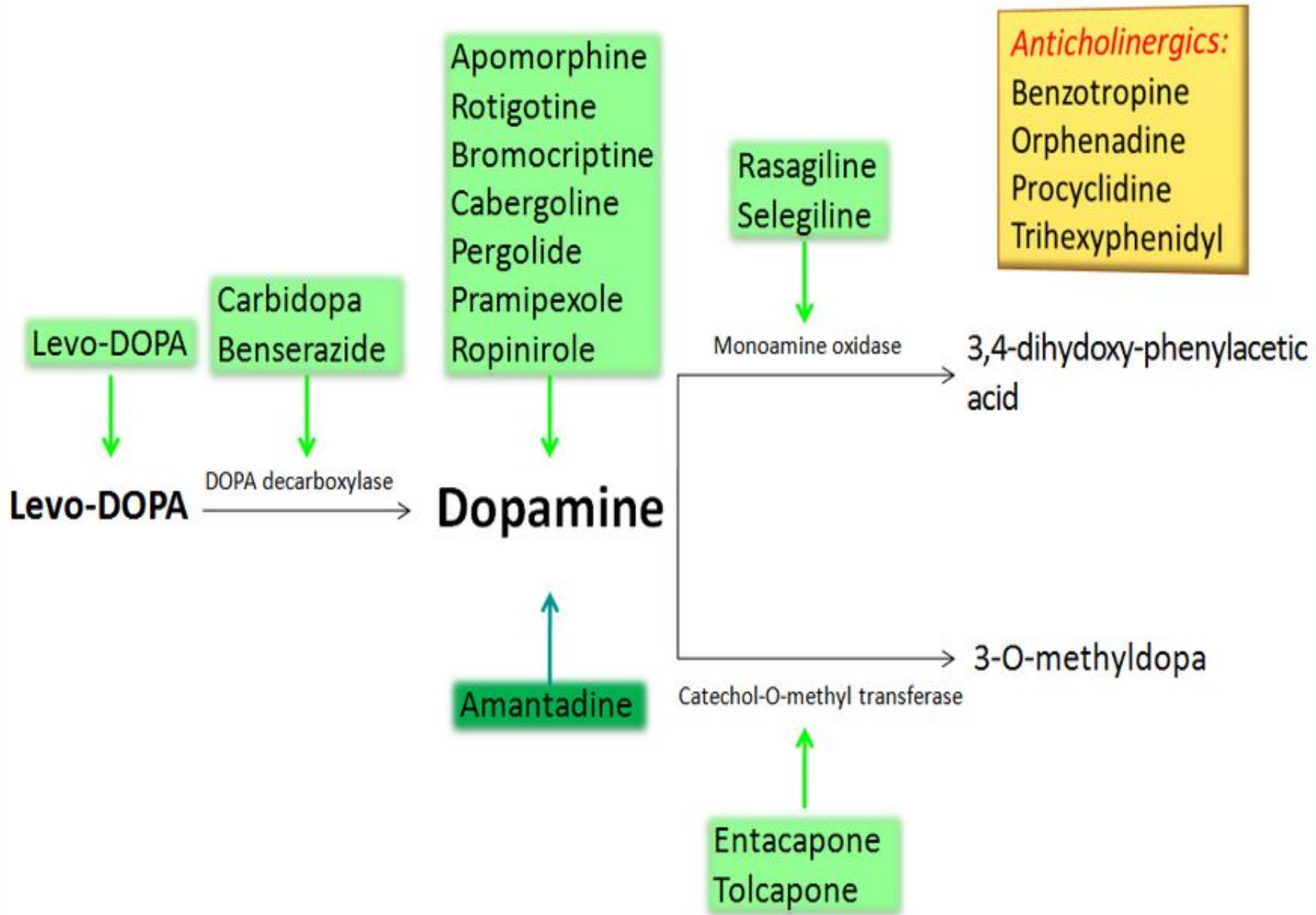
- Procyclidine - Trihexphenidyl

Anti-Parkinson Drugs —

- ◉ **Levodopa** → Dopamine precursor; converted to dopamine in brain.
- ◉ **Carbidopa** → Inhibits peripheral DOPA decarboxylase; saves levodopa.
- ◉ **Dopamine agonists** → Directly stimulate dopamine receptors.
- ◉ **Apomorphine** → Dopamine agonist; used for severe **OFF** periods.
- ◉ **MAO-B inhibitors** → Decrease dopamine breakdown.

- ◉ **COMT inhibitors** → Decrease levodopa metabolism; prolong its action.
- ◉ **Amantadine** → Increases dopaminergic activity + inhibits NMDA receptors.
- ◉ **Antimuscarinics** → Decrease muscarinic acetylcholine effects. 9.

ANTIPARKINSONISM



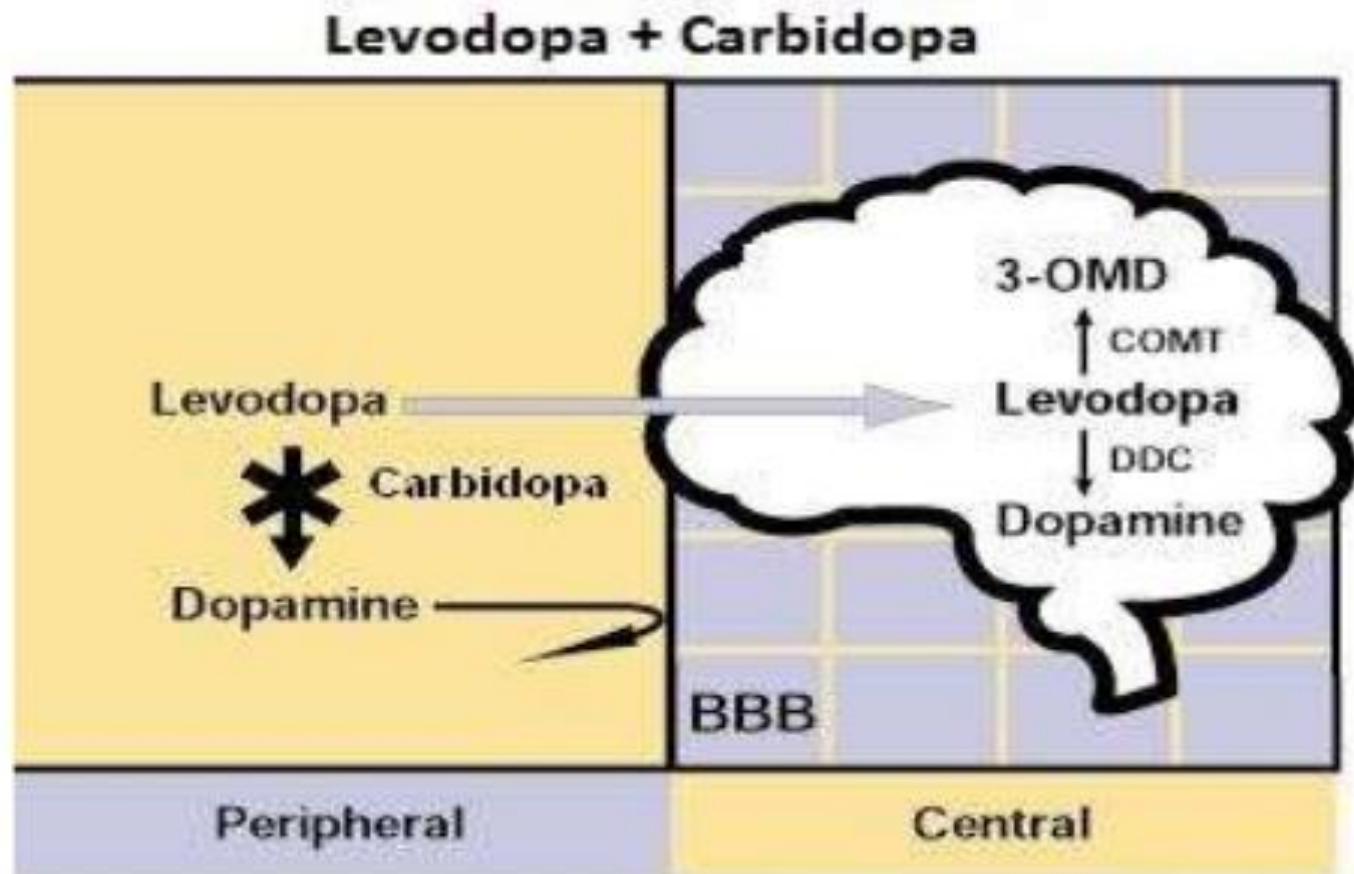
LEVODOPA

((-1)-3-(3,4-dihydroxyphenyl)-l-alanine)

Mechanism Of Action Of Levodopa

- Dopamine does not cross the blood-brain barrier**
- Levodopa, the immediate metabolic precursor of dopamine, does enter the brain (via an L-amino acid transporter, LA) where it is decarboxylated to dopamine.**

Mechanism of Action:



Pharmacokinetics Of Levodopa

- Rapidly absorbed from the small intestine**
- Ingestion of food delays the appearance of levodopa in the plasma**
- Half Life: 1 – 3 hrs**
- Main metabolic products**
 - * 3-methoxy-4-hydroxyphenyl acetic acid (homovanillic acid, HVA)**
 - * Dihydroxyphenylacetic acid (DOPAC).**

- 1-3% of administered levodopa actually enters the brain unaltered, the remainder is metabolized extra-cerebrally by decarboxylation to dopamine
- Concomitant administration of a peripheral dopa decarboxylase inhibitor such as carbidopa results in

***Peripheral metabolism of levodopa is reduced,**

***Plasma levels of levodopa are higher**

***Plasma half-life is longer,**

***More dopa is available for entry into the brain**

CLINICAL USES

Levodopa is effective in relieving bradykinesia and any disabilities resulting from Parkinsonism.

Combination Treatment

- Sinemet-25/100

(carbidopa 25 mg, levodopa 100 mg) three times daily

- Sinemet-25/250

(carbidopa 25 mg, levodopa 250 mg) three or four times daily

Adverse Effects

1) GIT

Anorexia, Nausea , Vomiting

2) CVS

Hypertension, Cardiac Arrhythmias

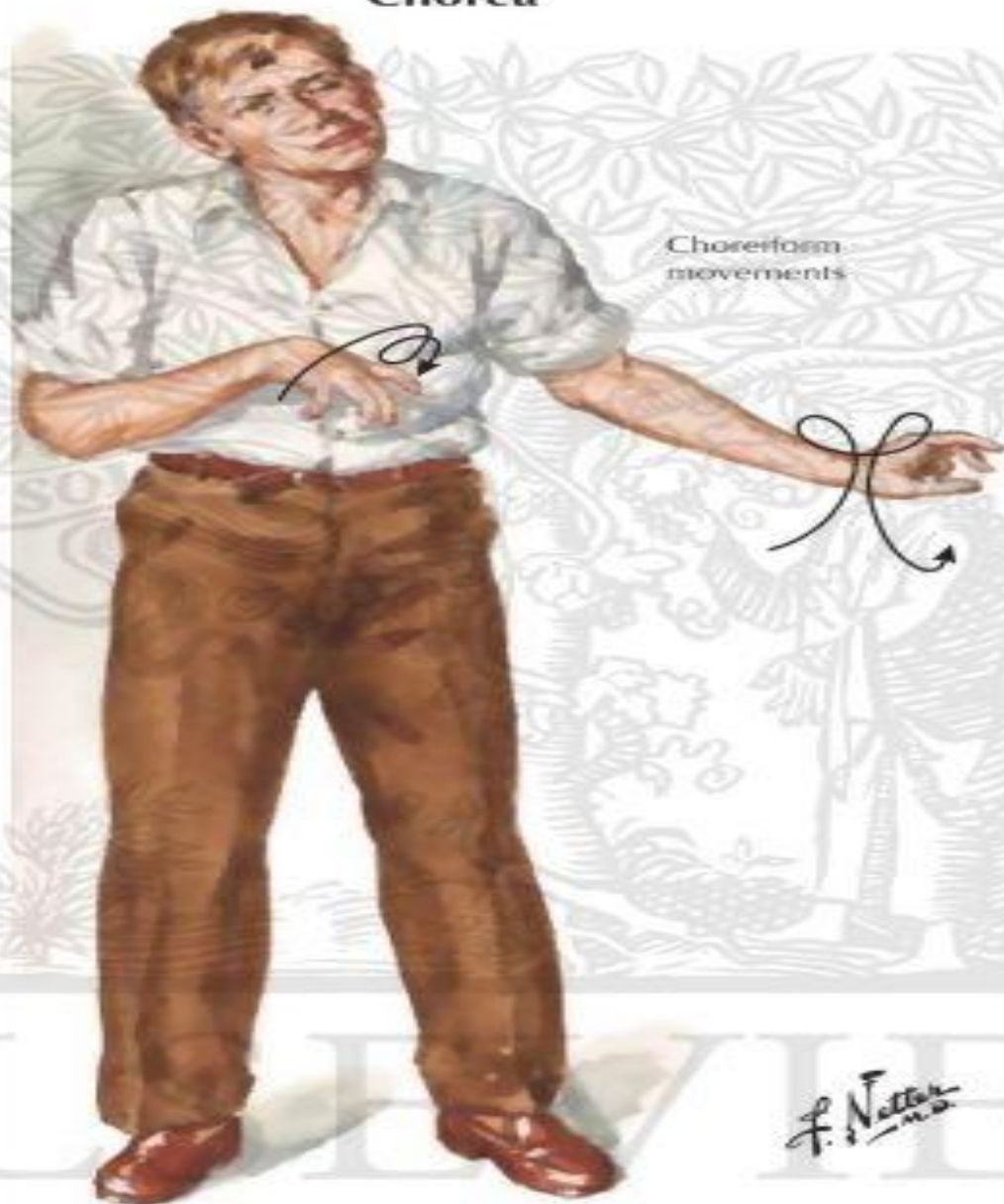
3) Dyskinesia

Choreoathetosis of the face and distal extremities

4) Behavioral Effects

Depression, Anxiety, Agitation, Insomnia, Somnolence, Confusion, Delusions, Hallucinations, Nightmares, Euphoria

Chorea



F. Netter M.D.

Levodopa Response Fluctuations

Wearing-off

End of dose → drug effect decreases →

Parkinson symptoms return

On-off phenomenon

Patient fluctuates between:

ON 😊 → **OFF** 🐢 → **ON** 😊 → **OFF** 🐢

Severe OFF period

➡ **Apomorphine**

5) Fluctuations In Response

- Wearing- Off reactions OR End-of-dose Akinesia
- On-Off Phenomenon
(patients with severe off-periods apomorphine is given)

6) Miscellaneous

- Mydriasis - hemolysis
- Precipitation of gout
- Brownish discoloration of saliva, urine or vaginal secretions

Drug Holidays

- Discontinuance of the drug for 3-21 days
- Temporarily improve responsiveness to levodopa
- Alleviate some of its adverse effects

Drug Interactions

- Pyridoxine (vitamin B₆)
- Monoamine oxidase A inhibitors
- Typical Antipsychotics

Contraindications

- Psychotic patients
- Angle-closure glaucoma
- Active peptic ulcer
- Malignant melanoma
- Cardiac disease

Dopamine Receptor Agonist

- **Older dopamine agonists**

Bromocriptine

Pergolide

ergot derivatives

- **Newer agents**

Pramipexole

Ropinirole

Difference B/W Levodopa & Dopamine Agonist

- * They do not require enzymatic conversion to an active metabolite.**
- * Have no potentially toxic metabolites.**
- * Do not compete with other substances for active transport into the blood and across the blood-brain barrier**

-
- **Drugs selectively affecting dopamine receptors may have more limited adverse effects than levodopa**

*** First-line therapy for Parkinson's disease.**

*** Lower incidence of the response fluctuations and dyskinesias .**

*** Given to patients with parkinsonism who are taking levodopa and who have end-of-dose akinesia or on-off phenomenon or are becoming resistant to treatment with levodopa.**

ERGOT DERIVATIVE

Bromocriptine:

- ☐ **D₂ agonist**
- ☐ **Hallucination, delirium, orthostatic hypotension are common**
- ☐ **Dyskinesia is less prominent**
- ☐ **Used to treat Parkinson's disease & hyperprolactinemia**
- ☐ **Has potential to cause pulmonary and retroperitoneal fibrosis**
- ☐ **Contraindicated in MI, Peripheral Vascular Disease**

NON ERGOT DERIVATIVES

- ☐ **Hallucination, Insomnia, Dizziness, Constipation orthostatic hypotension are common**
- ☐ **Dyskinesia less frequent**
- ☐ **Do not exacerbate PVD**
- ☐ **Do not cause fibrosis**

Pramipexole & Ropinirole

- ❑ **Orally active**

Rotigotine

- ❑ **Transdermal delivery system**
- ❑ **Used to treat Motor deficits in those who never used levodopa / advanced parkinson's disease**

Apomorphine

- ❑ **Injectable**
- ❑ **Used for acute management of off phenomenon**

MONOAMINE OXIDASE INHIBITORS

Two types of monoamine oxidase in CNS

- Monoamine oxidase A: Metabolizes norepinephrine and serotonin**
- Monoamine oxidase B: Metabolizes dopamine**

Selegiline

- * Selective irreversible inhibitor of monoamine oxidase B at normal doses**
- * Prolongs the antiparkinsonism effect of levodopa**
- * Allowing the dose of levodopa to be reduced**
- * Used as adjunctive therapy for patients with a declining or fluctuating response to levodopa**

Inhibition of dopa decarboxylase



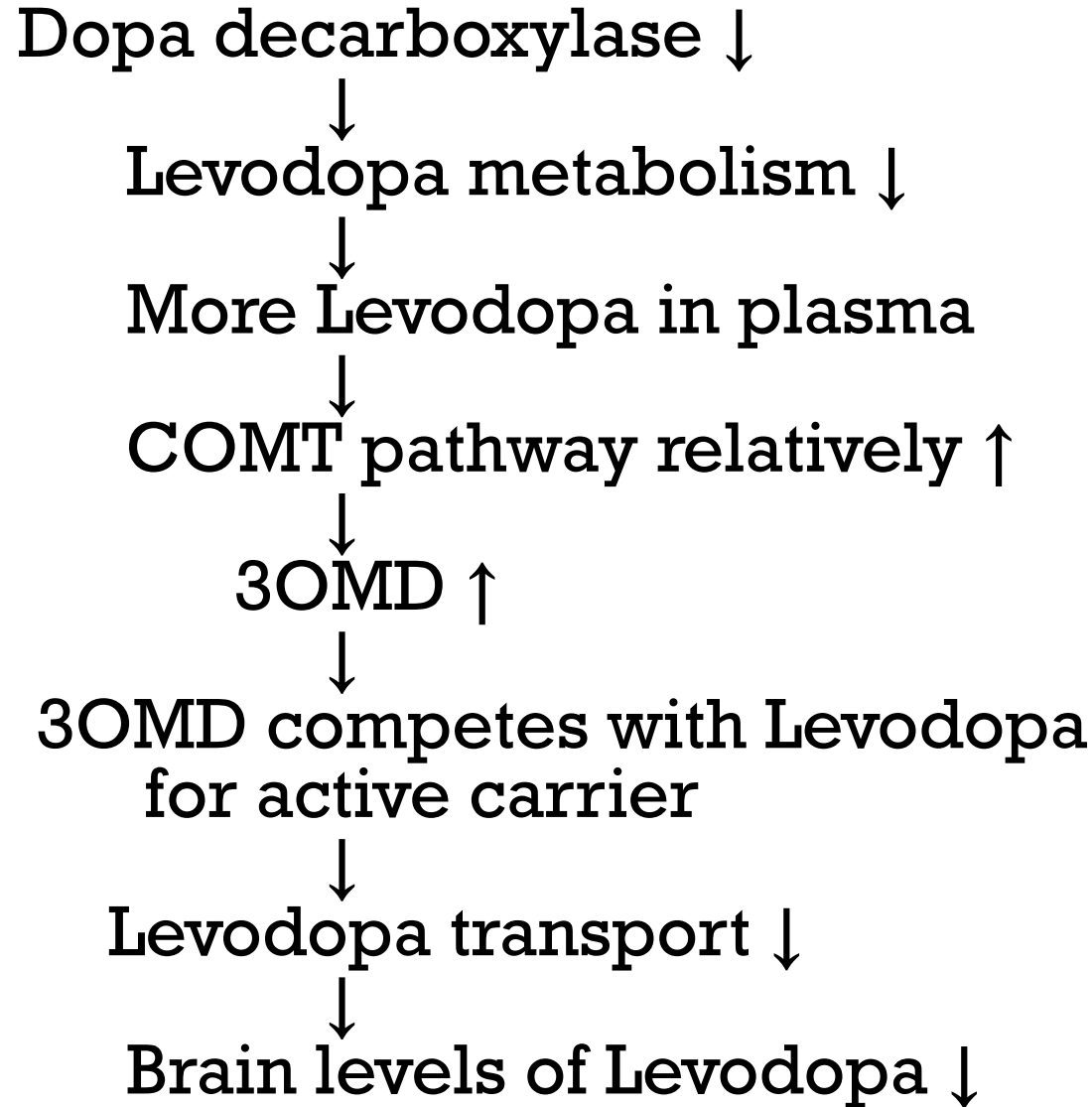
Activation of catechol-*O*-methyltransferase (COMT)



Increases plasma levels of 3-*O*-methyldopa (3OMD)



3OMD competes with levodopa for an active carrier mechanism



CATECHOL-O-METHYLTRANSFERASE INHIBITORS

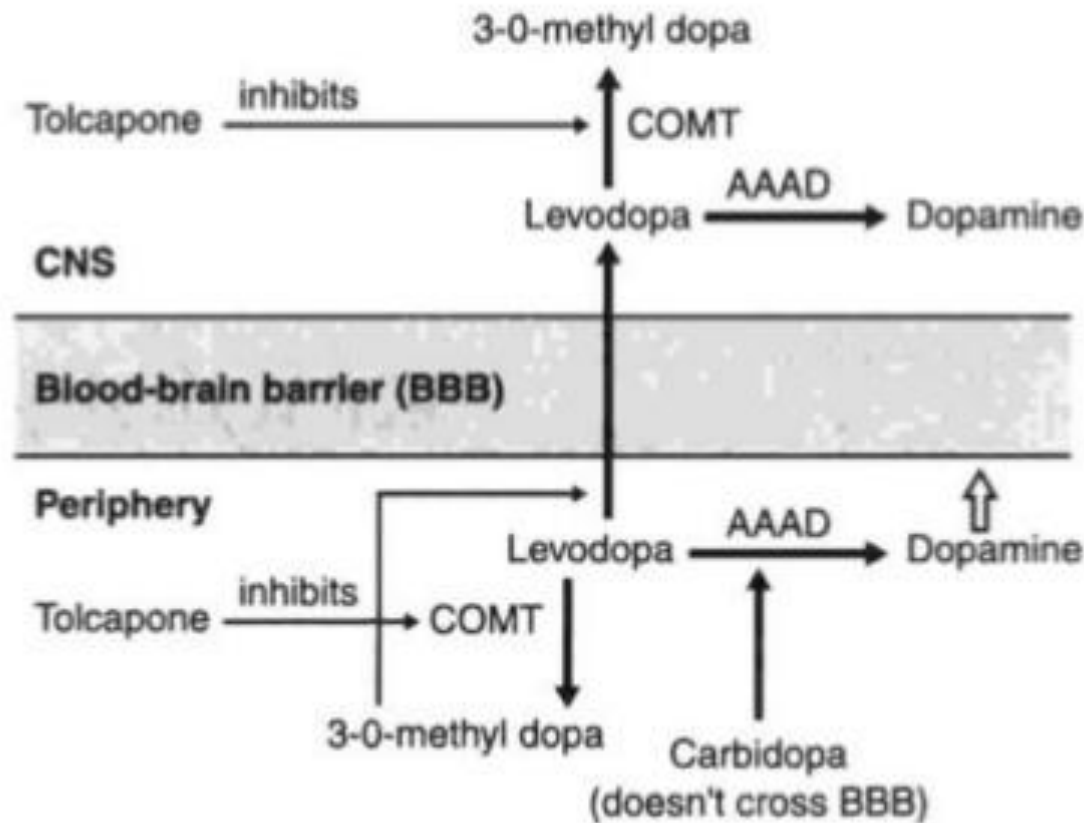
Tolcapone and Entacapone

Mechanism Of Action

- * Inhibiting catechol-O-methyltransferase (COMT)**
- * Prolong the action of levodopa by diminishing its peripheral metabolism**
- * Levodopa clearance is decreased,**
- * Bioavailability of levodopa is thus increased**

MECHANISM OF ACTIONS

Inhibition of COMT by entacapone or tolcapone leads to decreased plasma concentration of 3-O-methyldopa, increased central uptake of levodopa, and increased concentrations of brain dopamine.



AMANTADINE

- ❑ **ANTIVIRAL**
- ❑ **ANTIPARKINSONISM**
- ❑ **LESS EFFICACIOUS THAN LEVODOPA**
- ❑ **TOLERANCE DEVELOP MORE READILY**

MOA

- ❑ **INCREASES DOPAMINE RELEASE**
- ❑ **BLOCKING CHOLINERGIC RECEPTORS**
- ❑ **INHIBITING NMDA RECEPTORS**

AMANTADINE

ADRS

- ❑ **HIGH DOSES ACUTE TOXIC PSYCHOSIS**
- ❑ **ORTHOSTATIC HYPOTENSION**
- ❑ **URINARY RETENTION**
- ❑ **DRY MOUTH**
- ❑ **PERIPHERAL EDEMA**

1. The major neurotransmitter deficiency in Parkinsonism is:

- A. Acetylcholine
- B. Dopamine
- C. GABA
- D. Serotonin

2. Which drug is the metabolic precursor of dopamine used in Parkinsonism?

- A. Carbidopa
- B. Levodopa
- C. Bromocriptine
- D. Amantadine

3. Dopamine itself is not useful orally in Parkinsonism because it:

- A. Is rapidly excreted by kidney
- B. Cannot cross the blood-brain barrier
- C. Cannot bind dopamine receptors
- D. Causes severe hypoglycemia

4. Which enzyme metabolizes dopamine in the brain and is selectively inhibited by selegiline?

A. COMT

B. MAO-B

C. Dopa decarboxylase

D. Acetylcholinesterase

5. Which of the following is a COMT inhibitor?

A. Selegiline

B. Bromocriptine

C. Entacapone

D. Benztropine

6. A patient develops worsening Parkinsonian symptoms just before the next levodopa dose. This is called:

- A. On-off phenomenon
- B. Wearing-off phenomenon
- C. Dyskinesia
- D. Mydriasis

7. Severe OFF periods in patients receiving levodopa may be treated with:

- A. Apomorphine
- B. Carbidopa
- C. Entacapone
- D. Benztropine

8. Which of the following is a dopamine receptor agonist?

- A. Pramipexole
- B. Entacapone
- C. Selegiline
- D. Carbidopa

Why levodopa is given with Carbidopa?

Normally in peripheral tissues :

Levodopa → Dopamine

Dopa decarboxylase converts it.

Carbidopa inhibits peripheral dopa decarboxylase.

Result:

More levodopa → brain → dopamine